OM M. PATEL

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Vector AI Scholar at York University (MSc CS, AI specialization), with an interested in NLP and Optimization problems. I've built LLM systems for retrieval-grounded QA(RAG, agents) and research pipelines for constrained scheduling (JSSP). Based in Toronto and **eligible to work in Canada**.

EDUCATION

Master of Science in Computer Science, with AI Specialization
York University, Ontario, Canada **Sept 2025 – April 2027**

Relevant Courses: Data Mining, Probabilistic Model and Machine Learning, Fairness and Bias in AI, Machine Learning Theory, Robotics, Data Analysis and Visualization.

Bachelor of Technology in Computer Engineering (GPA: 3.47/4)
Pandit Deendayal Energy University, Gandhinagar, India **Nov 2021 – May 2025**

Relevant Courses: Computer Vision, Digital Image Processing, Pattern Recognition, Information Retrieval, Natural Language Processing, Big Data Analytics,

SKILLS

Languages	Python, SQL, JavaScript, HTML/CSS
ML/NLP & LLM	PyTorch, TensorFlow, Scikit-learn, HuggingFace (Transformers), LangChain, spaCy, NumPy, Pandas
Optimization & Graph ML	PyTorch Geometric, LoRA fine-tuning (4-bit), Constrained Prompting
Data & Backend	Airflow, Flask, MongoDB, GeoPandas
DevOps & Tools	Docker, CI/CD, AWS EC2, Azure, Streamlit, GitHub, Jira, Postman

RESEARCH & PUBLICATIONS

ADCIS 2024 (Springer): Automated Sleep Stage Classification using Machine Intelligence Techniques **Sept 2024 – Jul 2025**
Co-author

- Surveyed **90+** automated sleep staging studies and organized signal modalities (**EEG/ECG/EOG/EMG/**) and PSG data representations into an end-to-end taxonomy used for system design.
- Consolidated model families and evaluation practice (classical ML → deep learning) with clear reporting guidance around leakage, imbalance, and cross-subject generalization.

LLMs for Optimization : Job Shop Scheduling Optimization (JSSP) **Sept 2025 – Dec 2025**
Course Research Project (York University)

- Built a **GNN-augmented LLM pipeline** for scheduling by converting JSSP instances into disjunctive graphs (**PyTorch Geometric**) and serializing constraints into structured prompts.
- Pre-processed the noisy **STARJOB** dataset (~130k to **9,525** usable instances), reducing fine-tuning runtime from **70h to 11.5h** using **4-bit LoRA** (Unsloth) on an NVIDIA **RTX A6000**.

INDUSTRY EXPERIENCE

Backend & Data Engineering Intern **Dec 2024 – Aug 2025**
Uarra (Prev. Sharperly), Nigeria(Remote)

- Designed Airflow-powered **ETL pipelines** for geospatial datasets (GeoPandas, Python), increasing throughput by around **25%**.
- Deployed Docker-containerized services on Render and Azure via CI/CD pipelines.
- Optimized **MongoDB** schemas and indexing for real-time lookups used by downstream ML components.
- Collaborated with teams to implement ML models into production APIs and iteratively improved algorithm performance post-deployment.

Data Science Intern **Mar 2024 – July 2024**
Uarra, Nigeria(Remote)

- Developed geocoding APIs with **Python** (**Flask**, **Geopandas**) to reduce dependency on GMaps API cutting costs by up to **45%** (approx 350 USD) monthly.
- Scraped and pre-processed **20k+** addresses to create robust training datasets, producing versioned datasets for ML training.
- Trained and evaluated models (RNNs, Autoencoders, K-Means) for user behaviour prediction, route clustering, and anomaly detection

PROJECTS

GreenArm : *Vision Guided Robotic Waste Sorting* **Sept 2025 – Dec 2025**

- Co-led a 5-member team to deliver **ROS2** based **perception-to-manipulation system** to detect, classify, and pick waste using **YOLOv8**.
- Integrated **OpenCV-based** pixel-to-robot coordinate transformation for the Kinova Gen3 arm.
- Implemented **ArUco-based workspace calibration** with cached transforms, improving positioning accuracy to **0.5-1cm** precision under a fixed camera setup.

Legal Clarity : *Simplifying Legal Documents* **Aug 2024 – Nov 2024**

- Fine-tuned transformer models (**Pegasus**, **T5**, **IndicBARTSS**) in **PyTorch** to summarize and translate long-form, multilingual legal text for faster review.
- Packaged the pipeline into a **Flask** service for repeatable inference and downstream integration.

ChatPdf AI : *Chat with Your PDF* **Dec 2023 – Feb 2024**

- Built a retrieval-first document QA system using **LangChain** + **FAISS** to ground answers in source text.
- Integrated **OpenAI Agents SDK** for agentic workflows in document querying and response generation, resulting in **80%** accuracy on manual test set($N \leq 50$). Deployed on **Streamlit**.

Autograder : *Precision in Every Grade* **Sept 2023 – Nov 2023**

- Led a 6-person team to build an auto-grading pipeline (BERT-uncased, SpaCy, PyTesseract) with React front-end and Flask backend, enabling semi-automated grading for handwritten and typed responses

CERTIFICATIONS

- **Generative AI with LLMs** : Coursera
- **LangChain: Chat with your Data** : Coursera Project
- **LLMs: RAG with LlamaIndex & Azure OpenAI** : CredsVerse
- **Applications of AI/ML in Biomedical Signal Processing & CV** : PDEU
- **Understanding Incubation & Entrepreneurship** : NPTEL

EXTRACURRICULARS

- Vector AI Scholarship Recipient, York University — May 2025
- Top-10 rank Indian weather prediction at Kaggle ML Olympiad — April 2024 – May 2024
- AI/ML Mentor: Mentored 10+ students on projects including Autograder and Legal Clarity
- Volunteered at Vardaan Foundation: Taught essential digital literacy to empowering underprivileged students — June 2022 – Jul 2022